12/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations
Regd. Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph. 011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 9

Topics covered :

Physics : Ray Optics & Optical Instruments, Wave Optics

Chemistry : Alcohols, Phenols & Ethers, Aldehydes, Ketones & Carboxylic Acids

Botany : Microbes in Human Welfare, Organisms and Populations

Zoology : Human health & Disease, Strategies for Enhancement in Food Production: Animal husbandry

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.

(ii) Mark should be dark and should completely fill the circle.

(iii) Dark only one circle for each entry.

(iv) Dark the circle in the space provided only.

(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.

(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. A point object is placed at the centre of glass sphere of radius 6 cm and refractive index 1.5. The distance of the image from the surface of sphere is

   (1) 2 cm (2) 6 cm

   (3) 12 cm (4) 15 cm

2. What is distance of needle from a concave mirror of focal length 10 cm for which a virtual image of twice its height is formed?

   (1) 2.5 cm (2) 5 cm

   (3) 6 cm (4) 10 cm

3. Two thin lenses are in contact and focal length of combination is 80 cm. If the focal length of one lens is 20 cm, the power of other lens will be

   (1) 1.66 D (2) 4.00 D

   (3) –1.00 D (4) –3.75 D

4. A convex lens, a slab, a prism and a spherical solid ball has been prepared from same optically transparent material. Dispersive power will be possessed by

   (1) Glass slab and prism

   (2) The prism

   (3) The lens and solid ball

   (4) All the four

5. Two plano convex lenses of radius of curvature R and refractive index $\mu = 1.5$ will have equivalent focal length equal to R, when they are placed

   (1) At distance $R/4$

   (2) At distance $R/2$

   (3) At distance R

   (4) In contact with each other

6. The refractive index of water is 4/3 and that of glass is 5/3. What will be critical angle of ray of light entering water from glass?

   (1) $\sin^{-1}(4/5)$ (2) $\sin^{-1}(1/4)$

   (3) $\sin^{-1}(1/2)$ (4) $\sin^{-1}(3/4)$

7. A ray of light passes through an equilateral prism such that the angle of incidence is equal to angle of emergence and later is equal to ¾ th the angle of prism. The angle of deviation is
(1) 45° (2) 30°
(3) 20° (4) 37°

8. A disc is placed on the surface of pond which has refractive index 5/3. A source of light is placed 4 m below the surface of liquid. The minimum radius of disc needed so that light is not coming out, is
(1) 6 m (2) 4 m
(3) 3 m (4) 1.5 m

9. A thin prism P_1 with angle 5° and made from glass of refractive index 1.54 is combined with another prism P_2 made of glass of refractive index 1.92 to produce dispersion without deviation. The angle of prism P_2 is
(1) 5.33° (2) 2.9°
(3) 3.56° (4) 4.22°

10. A person cannot see the objects beyond 50 cm. The power of lens to correct her vision will be
(1) 2 D (2) –2 D
(3) + 4 D (4) –1 D

11. When wavelength of light used in an optical instrument are 4000 Å and 5000 Å. The ratio of respective resolving power is
(1) 4/5 (2) 5/4
(3) 16/25 (4) 25/16

12. When a ray of light enters a medium of refractive index 5/3, then it is observed that angle of refraction is half the angle of incidence. The value of angle of incidence is
(1) $2\sin^{-1}(5/6)$ (2) $2\cos^{-1}(5/6)$
(3) $2\cos^{-1}(6/5)$ (4) $2\sin^{-1}(6/5)$

13. A car is fitted with a convex side view mirror of focal length 20 cm. A second car 280 cm behind the first car is overtaking the first car at relative speed of 30 m/s. The speed of image of second car as seen in the mirror of first car is
(1) $2/5\ ms^{-1}$ (2) $2/3\ ms^{-1}$
(3) $2/15\ ms^{-1}$ (4) $1/15\ ms^{-1}$

14. An achromatic convergent doublet of two lenses in contact has a power of + 2D. The convex lens has focal length of 20 cm. What is ratio of dispersive powers of convergent and divergent lenses?
(1) 2 : 5
(2) 3 : 5
(3) 5 : 2
(4) 5 : 3

15. The resolution limit of eye is 1 minute. At a distance '*D*' km from eye, two persons stand with a lateral separation of 3 m. For two persons to be just resolved by naked eye, *D* should be nearly
(1) 10 km (2) 15 km
(3) 30 km (4) 25 km

16. An equi-convex lens of focal length 10 cm is silvered at one surface, new *F* will be (μ = 1.5)
(1) 2.0 cm (2) –40 cm
(3) +1.2 cm (4) –2.5 cm

17. A concave mirror has focal length of 20 cm. The distance between two positions of object for which image size is double of object size is
(1) 30 cm (2) 20 cm
(3) 40 cm (4) 60 cm

18. A circular beam of light of diameter 2 cm falls on a plane surface of glass. The angle of incidence is 60° and refractive index of glass is 1.5. The diameter of the refracted beam is
(1) 3.26 cm (2) 1.56 cm
(3) 4.21 cm (4) 1.76 cm

19. The magnification produced by an astronomical telescope for normal adjustment is 10 and length of telescope is 110 cm. The magnifying power when image is formed at least distance of distinct vision (D = 25 cm) is
(1) 10 (2) 12
(3) 8 (4) 14

20. When a convergent beam of light passes through a diverging lens, the result incorrect is
(1) It may be a convergent beam
(2) It may be a divergent beam
(3) It may be a parallel beam
(4) It must be a parallel beam

21. A microscope has an objective of focal length 1.5 cm and an eye piece of 2.5 cm. If the distance between objective and eye piece is 25 cm, what is the approximate value of magnification produced for relaxed eye?
(1) 120.5 (2) 130.6
(3) 140.6 (4) 170.2

22. The splitting of white light into several colours on passing through glass prism is due to
(1) Refraction (2) Reflection
(3) Interference (4) Dispersion

23. A small angle prism (μ = 1.62) gives a deviation of 4.8°. The angle of the prism is
(1) 3.42° (2) 7.74°
(3) 6.84° (4) 2.4°

24. Focal length of an equiconvex lens of refractive index 3/2 in air is 20 cm. What would be its focal length in water (μ = 4/3)?
(1) 60 cm (2) –80 cm
(3) +80 cm (4) –60 cm

25. An object of height 1.5 cm is placed on the axis of a convex lens of focal length 25 cm. A real image is formed at a distance of 75 cm from the lens. The size of image is
(1) 4.5 cm (2) 1.3 cm
(3) 2.6 cm (4) 3.0 cm

26. The magnifying power of a telescope can be increased by
(1) Increasing focal length of objective
(2) Increasing focal length of eye piece
(3) Fitting eye piece of higher power
(4) Both (1) and (3) are correct

27. Resolving power of a reflecting type of telescope increases with
(1) Increasing diameter of objective mirror
(2) Increasing the wavelength of incident ray
(3) Decreasing diameter of objective mirror
(4) Both (2) and (3) are correct

28. A bird flies down vertically towards water surface. To a fish inside water, vertically below the bird, the bird will appear
(1) Be closer than its actual distance
(2) Moves slower than its actual speed
(3) Farther away than its actual distance
(4) Moves at same speed as bird flies

29. Two objects are placed on principal axis of a thin converging lens. One is 10 cm from the lens and other is on the other side of the lens at 40 cm from lens. The image of both objects are in same plane. What is focal length of lens?
(1) 16 cm (2) 24 cm
(3) 20 cm (4) 32 cm

30. In Young's double slit experiment, if the distance between two slits is $d/3$, distance between screen and slits is $3D$ and light of wavelength is 3λ then fringe width will be
(1) $\frac{81\lambda d}{D}$ (2) $\frac{27\lambda D}{d}$
(3) $\frac{27\lambda d}{D}$ (4) $\frac{81\lambda D}{d}$

31. Two coherent interfering beams have intensity ratio 9 : 4. The intensity ratio of maximum to minimum on screen pattern
(1) 25 : 1 (2) 13 : 9
(3) 5 : 1 (4) 16 : 81

32. Two slits 4 mm apart are illuminated by light of wavelength 6000 Å. What will be fringe width on screen placed 2 m from the slits plane?
(1) 4.0 mm (2) 0.3 mm
(3) 0.05 mm (4) 0.6 mm

33. Polaroid glass is used in sun glasses because
(1) It has good colour
(2) It is easy to clean
(3) It reduces light intensity to half on account of polarisation
(4) It is having high refractive index

34. A star is emitting yellow light. If it is accelerated towards earth, then to an observer on earth, it will appear
(1) Red in colour
(2) Orange in colour
(3) Yellow in colour
(4) Gradually changing to blue

35. A light beam is incident at an angle of 53° on interface of air-glass such that angle between refracted and reflected beams is 90°. The refracting index of glass is
(1) 3/4 (2) 3/2
(3) 4/3 (4) 5/3

36. A single slit of width b is illuminated by light of wavelength 400 nm and width of central diffraction pattern is measured as y. When half of slit width is covered and illuminated by yellow light of wavelength 600 nm, the width of diffraction pattern is
(1) $2y$ (2) $3y$
(3) $y/2$ (4) $5/2y$

37. The angular resolution of a 10 cm diameter telescope at a wavelength of 5000 Å is of the order
(1) 10^{-4} rad (2) 10^{-5} rad
(3) 10^{-6} rad (4) 10^{-3} rad

38. A thin mica sheet of thickness 2×10^{-6} m and refractive index 1.5 is introduced in the path of one of the waves. The wavelength of the wave used is 5000 Å. The central bright maxima will shift by
(1) 2 fringes (2) 3 fringes
(3) 5 fringes (4) 6 fringes

39. In YDSE, the slits are 2 mm apart and are illuminated with mixture of two wavelengths λ_1 = 12000 Å and λ_2 = 10000 Å. At what minimum distance from common central bright fringe on screen 2 m from slits will a bright fringe from one interference pattern coincide with a bright fringe of other?
(1) 2.7 mm (2) 3.2 mm
(3) 4.6 mm (4) 6.0 mm

40. Two nicol prisms are oriented with their principal planes making an angle of 60°. The percentage of incident unpolarised light which passes through the system

(1) 25% (2) 37.5%

(3) 50% (4) 12.5%

41. A star is receding away from earth with a velocity of 10^5 m/s. If the wavelength of its spectral line is 5700 Å, then Doppler shift will be

(1) 2.5 Å (2) 3.5 Å

(3) 1.9 Å (4) 2.7 Å

42. Young's double slit experiment is performed in a liquid. The 10th bright fringe in liquid lies where 6th dark fringe lies in vacuum. The refractive index of liquid is approximately

(1) 1.6

(2) 1.7

(3) 1.9

(4) 1.8

43. A beam of electrons is used in YDSE. The slit width is *d*. If the velocity of the electron is increased

(1) Fringe width decreases

(2) Fringe width increases

(3) No interference pattern observed

(4) Whole fringe pattern shifts from original centre

44. A flat glass slab of thickness 6 cm and refractive index 1.5 is placed in front of a plane mirror. An observer is standing behind glass slab and looking at mirror. The actual distance of the observer from the mirror is 50 cm. The distance of his image from himself, as seen by observer is

(1) 94 cm (2) 96 cm

(3) 100 cm (4) 102 cm

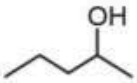
45. A concave mirror of focal length f_1 is placed at a distance of *d* from *a* convex lens of focal length f_2. A beam of light coming from infinity and falling on convex lens – concave mirror combination returns to infinity. Among the following the distance *d* must be equal to

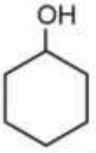
(1) $|f_1| + |f_2|$ (2) $|f_2| - |f_1|$

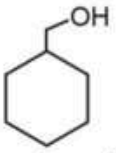
(3) $|f_2| - 2|f_1|$ (4) $2|f_1| + |f_2|$

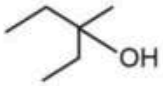
CHEMISTRY

46. The compound which will give immediate turbidity with concentrated HCl and $ZnCl_2$ is

(1) 

(2) 

(3) 

(4) 

47. The pair of compounds which can be distinguished by iodoform test is

(1) Acetone and 2-butanol

(2) Acetaldehyde and benzaldehyde

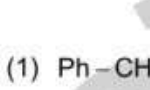
(3) Benzaldehyde and benzophenone

(4) 2-Methylpropanal and methanal

48. Consider the following reaction sequence

$$Ph-CH_2-CN \xrightarrow[(ii)\,H_3O^+]{(i)\,CH_3MgBr} A \xrightarrow[Conc.\,HCl]{Zn-Hg} B(Major)$$

Product B is

(1) 

(2) $Ph-CH_2-CH_2-CH_2-CH_3$

(3) $Ph-CH=CH-CH_3$

(4) $Ph-CH_2-CH_2-CH_2-OH$

49. In which of the given reactions C–C bond formation does not occur?

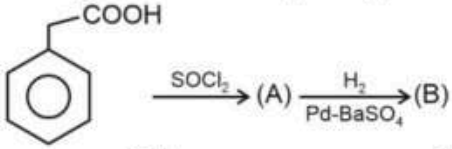
(1) Etard reaction

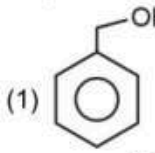
(2) Aldol condensation reaction

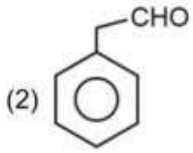
(3) Gattermann-Koch reaction

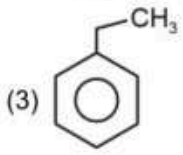
(4) Friedel-Crafts reaction

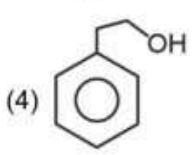
50. In the given reaction sequence, product-B is



(1) 

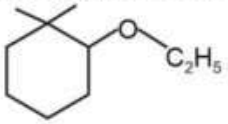
(2) 

(3) 

(4) 

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51. IUPAC name of the given compound is



(1) 1-Ethoxy-2,2-dimethylcyclohexane

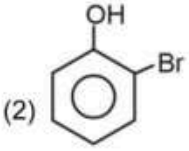
(2) 1-Ethoxy-6,6-dimethylcyclohexane

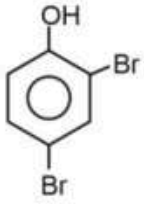
(3) 2-Ethoxy-1,1-dimethylcyclohexane

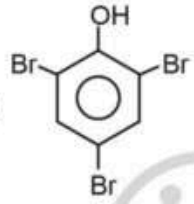
(4) 6-Ethoxy-1,1-dimethylcyclohexane

52. When phenol reacts with Br_2/water, the major product obtained is

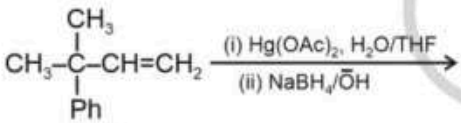
(1) 

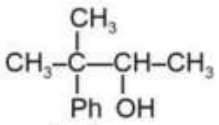
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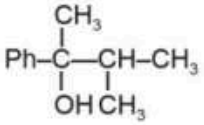
(3) 

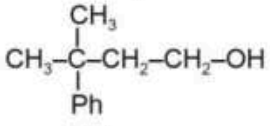
(4) 

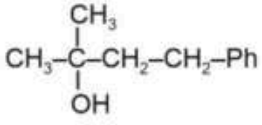
53. Major product of the given reaction is



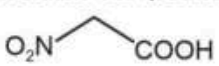
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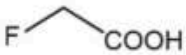
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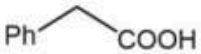
(3) 

(4) 

54. Most acidic compound among the following is

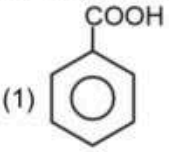
(1) 

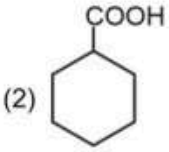
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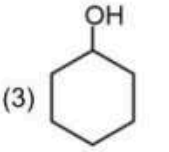
(3) 

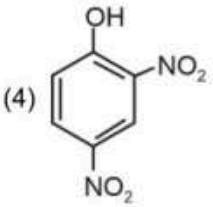
(4) HCOOH

55. The compound which does not liberate CO_2 on reaction with $NaHCO_3$?

(1) 

(2) 

(3) 

(4) 

56. The compound which does not give positive Fehling's test is

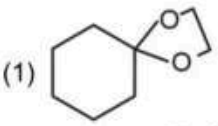
(1) Acetaldehyde

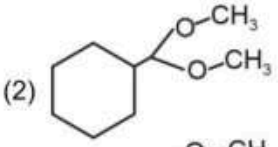
(2) Formaldehyde

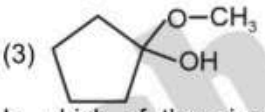
(3) Benzaldehyde

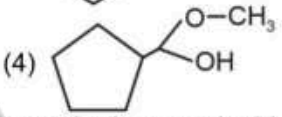
(4) Cyclohexanecarbaldehyde

57. Hemiketal among the following is

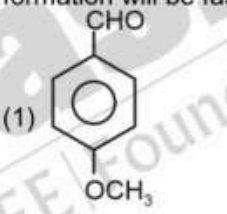
(1) 

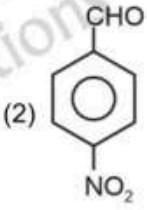
(2) 

(3) 

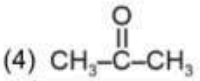
(4) 

58. In which of the given compounds cyanohydrin formation will be fastest?

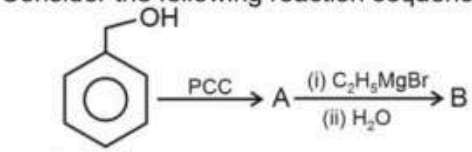
(1) 

(2) 

(3) CH_3CHO

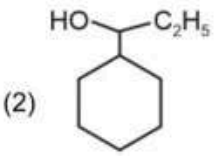
(4) 

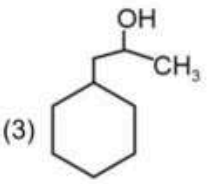
59. Consider the following reaction sequence,

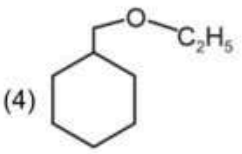


Product B is

(1) C_2H_6

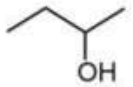
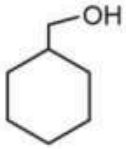
(2) 

(3) 

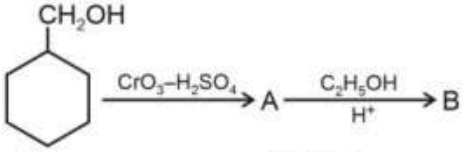
(4) 

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60. Which among the following alcohols when passed over heated copper at 573 K gives ketone as major product?

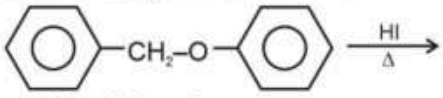
(1) 
(2) 
(3) 
(4) 

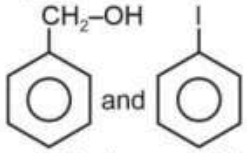
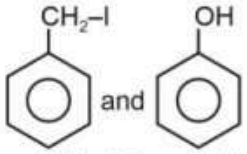
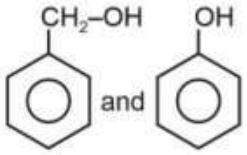
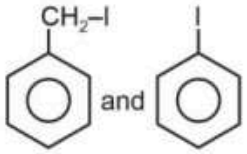
61. In the given reaction sequence, product B is



(1) Ether (2) Ester
(3) Acetal (4) Ketone

62. Products of the given reaction are



(1) 
(2) 
(3) 
(4) 

63. Correct order of acidity of the given alcohols is

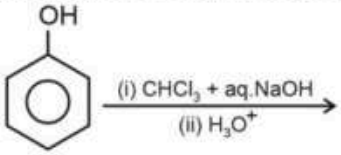
$(CH_3)_3C–OH$ $CH_3CH_2–OH$ $(CH_3)_2CH–OH$
(a) (b) (c)

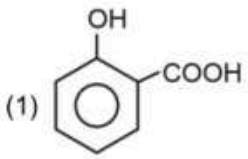
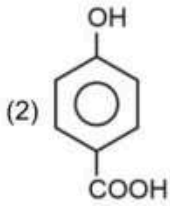
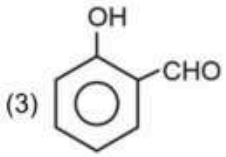
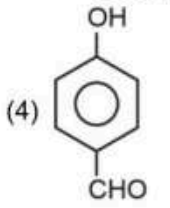
(1) (a) > (c) > (b) (2) (b) > (c) > (a)
(3) (a) > (b) > (c) (4) (c) > (a) > (b)

64. Cumene hydroperoxide on reaction with dilute H_2SO_4 gives

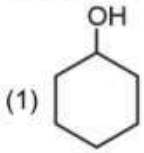
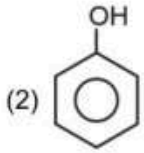
(1) Benzaldehyde and ethanol
(2) Phenol and acetone
(3) Acetophenone and methanol
(4) Benzyl alcohol and acetaldehyde

65. In the given reaction, major product formed is

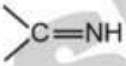
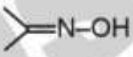


(1) 
(2) 
(3) 
(4) 

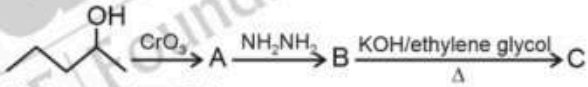
66. The compound which does not react with sodium metal is

(1) 
(2) 
(3) $CH_3CH_2OCH_2CH_3$ (4) $CH_3CH_2–COOH$

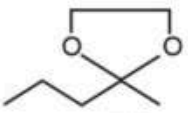
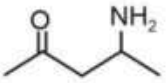
67. Which carbonyl derivative is called oxime?

(1) 
(2) 
(3) 
(4) 

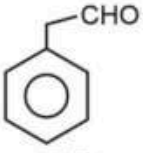
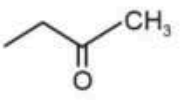
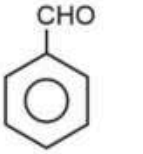
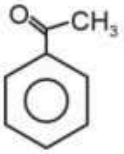
68. Consider the following reaction sequence



Major product C is

(1) 
(2) 
(3) 
(4) 

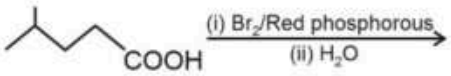
69. The compound which will not give aldol condensation reaction is

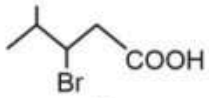
(1) 
(2) 
(3) 
(4) 

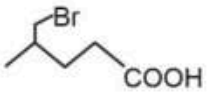
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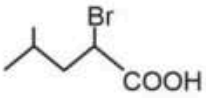
70. Benzaldehyde when heated with concentrated NaOH gives
(1) Benzyl alcohol and sodium benzoate
(2) Phenol and sodium benzoate
(3) Toluene and sodium benzoate
(4) Benzyl benzoate

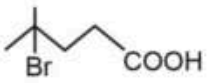
71. Major product of the given reaction is

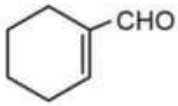


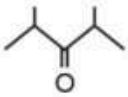
(1) 

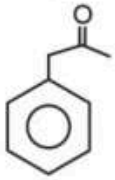
(2) 

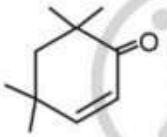
(3) 

(4) 

72. Compound which does not show keto enol tautomerism is
(1) 

(2) 

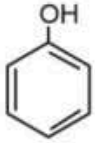
(3) 

(4) 

73. Which among the following is an allyl alcohol?
(1) 

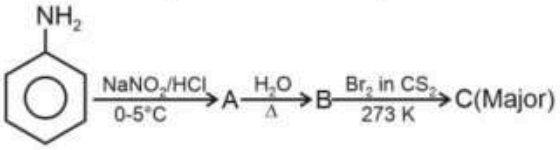
(2) 

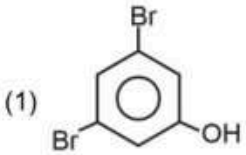
(3) 

(4) 

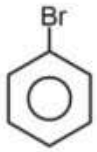
74. Electrophile in Kolbe's reaction is
(1) $:CCl_2$ (2) CO_2
(3) $H-\overset{+}{C}=O$ (4) $CH_3-\overset{+}{C}=O$

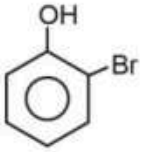
75. Consider the given reaction sequence

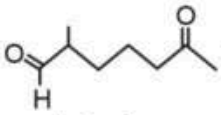


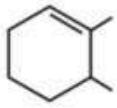
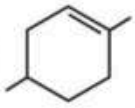
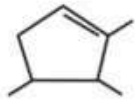
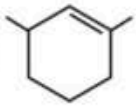
Major product C is
(1) 

(2) 

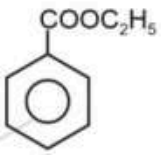
(3) 

(4) 

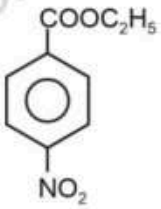
76. Compound 

is obtained by the reductive ozonolysis of
(1) 
(2) 
(3) 
(4) 

77. The compound which is hydrolysed most easily in alkaline medium is
(1) 

(2) 

(3) 

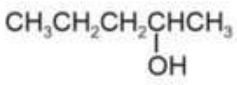
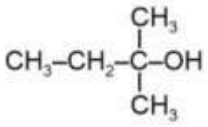
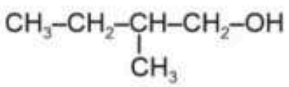
(4) 

78. Consider the following statements
(a) Grignard reagents react with dry ice to form salts of carboxylic acids.
(b) Benzamide on acidic hydrolysis gives phenylacetic acid.
(c) Cumene is oxidised to benzoic acid when heated with alkaline $KMnO_4$ solution followed by acidic work-up.

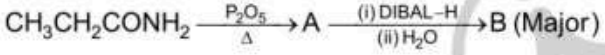
The correct statements are
(1) (a) and (b) only
(2) (b) and (c) only
(3) (a) and (c) only
(4) (a), (b) and (c)

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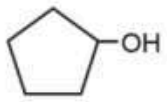
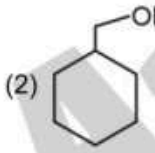
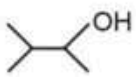
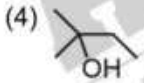
79. Incorrect statement among the following is
(1) Methanol is known as wood spirit
(2) Enzyme zymase converts glucose to ethanol
(3) Fermentation of glucose takes place in anaerobic conditions
(4) Commercial ethanol is made unfit for drinking by mixing in it copper chloride and aniline

80. The alcohol which is dehydrated most easily by concentrated sulphuric acid is
(1) $CH_3CH_2CH_2CH_2CH_2–OH$
(2) 
(3) 
(4) 

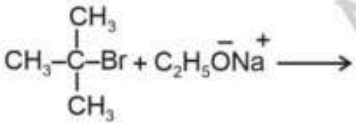
81. In the given reaction sequence, major product B is

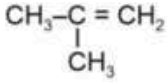
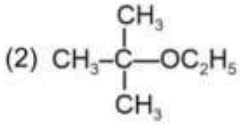
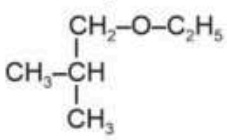


(1) $CH_3CH_2CH_2NH_2$ (2) CH_3CH_2CHO
(3) $CH_3CH_2CH_2OH$ (4) $CH_3CH_2CH_3$

82. The compound which gives red colour in Victor Meyer's test is
(1) 
(2) 
(3) 
(4) 

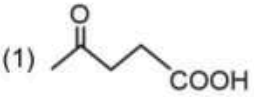
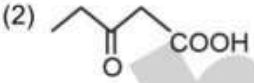
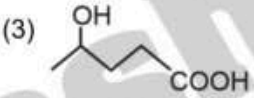
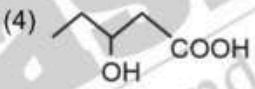
83. In the given reaction, major product formed is



(1) 
(2) 
(3) 
(4) $CH_3–CH=CH–CH_3$

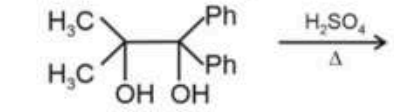
84. Compound which will not react with Tollen's reagent is
(1) Propanal
(2) Formic acid
(3) Formaldehyde
(4) 2-Butanone

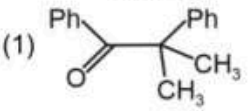
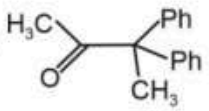
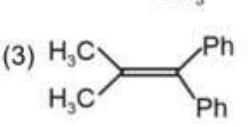
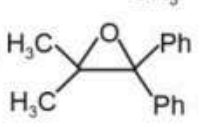
85. Oxidation of toluene with CrO_3 in presence of $(CH_3CO)_2O$ followed by acidic hydrolysis gives product A, which on treatment with phenylhydrazine gives
(1) $PhCH = N – NH – Ph$
(2) $PhCH_2NH – NH – Ph$
(3) $PhCH_2 – NH – NH_2$
(4) $PhCH = N – NH_2$

86. Which among the following undergoes decarboxylation most easily on heating?
(1) 
(2) 
(3) 
(4) 

87. Incorrect statement among the following is
(1) Methanal is a gas at room temperature
(2) Boiling point of propanal is higher than methoxy ethane
(3) Boiling point of acetone is higher than ethanol
(4) $NaHSO_3$ is a useful reagent for separation and purification of aldehydes

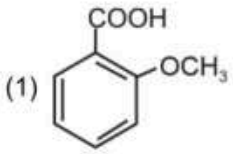
88. Major product of the given reaction is

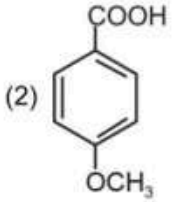


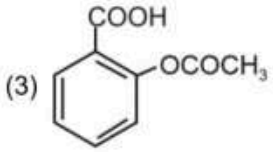
(1) 
(2) 
(3) 
(4) 

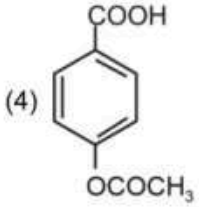
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89. Structure of Aspirin is

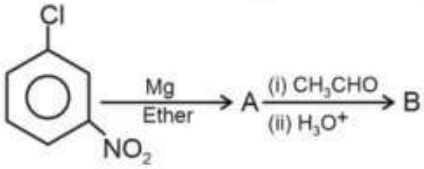
(1) 

(2) 

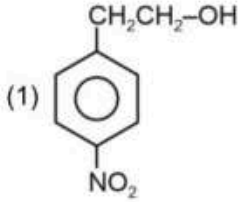
(3) 

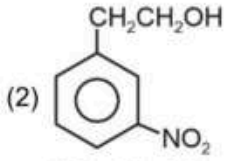
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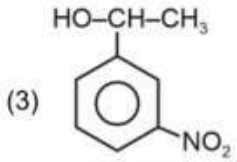
90. Consider the following reaction sequence

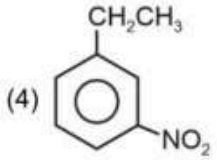


Product B is

(1) 

(2) 

(3) 

(4) 

BOTANY

91. All of the following can be grown in cell free culture medium, **except**
    (1) Fungi
    (2) Viruses
    (3) Bacteria
    (4) Protozoans

92. Curd is more nutritious than milk because curd has
    (1) Carbohydrates such as lactose and glucose
    (2) Large amount of proteins
    (3) More vitamins and beneficial microbes
    (4) Various types of mineral nutrients

93. Microorganism that converts milk into curd
    (1) Produce acids that coagulate milk proteins
    (2) Is a unicellular nucleated organism
    (3) Is an autotrophic organism
    (4) Does not multiply by binary fission

94. One of the oldest food items in which microbes were used is
    (1) Cheese
    (2) Dosa
    (3) Coconut milk
    (4) Idli

95. Swiss cheese and Roquefort cheese are ripened with the help of a
    (1) Fungus and a bacterium respectively
    (2) Bacterium and a fungus respectively
    (3) Cyanobacterium and fungus respectively
    (4) Bacterium

96. Fermenting agent used in preparing dough for making bread is
    (1) *Saccharomyces cerevisiae*
    (2) *Pseudomonas cepacia*
    (3) *Aspergillus fumigatus*
    (4) *Streptococcus lactis*

97. Puffed-up appearance of dough during fermentation is due to production of
    (1) Ethyl alcohol
    (2) Lactic acid
    (3) Water vapour
    (4) Carbon dioxide

98. Toddy is made by fermenting sap from
    (1) Maize plants
    (2) Barley plants
    (3) Palms
    (4) *Pinus* trees

99. Select the **incorrect** statement from the following.
(1) Same fermenting agent can be used in making bread and beer
(2) Wine and beer both are produced without distillation
(3) Brandy has less percentage of alcohol than beer
(4) The microbes used for fermenting fruit juices to produce alcohol are called brewer's yeast

100. Antibiotics means against life. This statement is in the context of
(1) Human beings
(2) Disease causing organisms
(3) All living organisms
(4) Animal beings only

101. The first antibiotic that was discovered is
(1) Streptomycin (2) Erythromycin
(3) Tetracycline (4) Penicillin

102. For which of the following works Ernest Chain and Howard Florey were awarded the Nobel prize?
(1) Establishment of full potential of penicillin as an effective antibiotic
(2) Use of microbes for the production of antibiotics
(3) Finding out the molecular structure of streptomycin
(4) Finding out the technique for the formation of penicillin synthetically

103. While working on a particular bacterium, Alexander Fleming observed that growth of this bacterium was retarded in the presence of a mould. That particular bacterium was
(1) *Staphylococcus*
(2) *Clostridium*
(3) *Leuconostoc*
(4) *Streptococcus*

104. Select the **incorrect** match w.r.t. microbes and their commercial products.

(1)	*Clostridium butylicum*	–	Gluconic acid
(2)	*Aspergillus niger*	–	Citric acid
(3)	*Acetobacter aceti*	–	Acetic acid
(4)	*Lactobacillus*	–	Lactic acid

105. Which enzyme produced by *Geotrichum candidum* is used in detergent formulation that removes oil strains from laundry?
(1) Amylase (2) Maltase
(3) Lipase (4) Trypsin

106. Identify the statement which is **true** for the microbes used in the production of cyclosporin A and statins respectively.
(1) Both the microorganisms belong to the same class of fungi
(2) Both are unicellular organisms
(3) One of them is a bacterium whereas the other is a fungus
(4) Both of them reproduce asexually

107. Read the following statements :
(a) It is produced by a bacterium.
(b) It is used as an immunosuppressive agent.
(c) It is useful for patients who have undergone myocardial infraction.
(d) It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.

Which amongst the given statements is/are **true** for streptokinase?
(1) (a) and (d) (2) (a) and (c)
(3) (b) and (d) (4) (c) only

108. In sewage treatment, the primary effluent is sent to
(1) Anaerobic sludge digester
(2) Natural water bodies
(3) Aeration tank for secondary treatment
(4) Irrigate crop fields

109. Biological treatment of sewage water is done with the help of
(1) Photosynthetic green algae
(2) Aquatic plants and animals
(3) Anaerobic and aerobic microorganisms
(4) Aerobic microorganisms only

110. In STPs, secondary treatment is done to
(1) Increase BOD of waste water
(2) Reduce the amount of organic waste in water
(3) Reduce the amount of dissolved oxygen in waste water
(4) Remove pesticides, pathogens and turbidity in waste water

111. Methanogenic bacteria digest other bacteria and fungi and produce gases such as methane and carbon dioxide. During sewage treatment this event occurs
(1) In anaerobic sludge digester
(2) During tertiary treatment
(3) In primary settling tank
(4) In filtration tank

112. The major component of biogas is
(1) Carbon dioxide
(2) Methane
(3) Hydrogen
(4) Butane

113. In organic farming,
(1) Both biopesticides and chemical pesticides are used in equal amount
(2) Only wild varieties of plants are produced
(3) Plants are grown only in a closed environment
(4) Pests are kept at manageable levels

114. Match the following columns and select the **correct** option.

	Column-I		**Column-II**
a.	*Trichoderma*	(i)	Controls butterfly caterpillars
b.	*Bacillus thuringiensis*	(ii)	Effective against several plant pathogens
c.	Ladybird beetle	(iii)	Pathogen that attacks specific insect
d.	Baculovirus	(iv)	Controls aphids

(1) a(iii), b(iv), c(i), d(ii)
(2) a(ii), b(i), c(iii), d(iv)
(3) a(iii), b(i), c(iv), d(ii)
(4) a(ii), b(i), c(iv), d(iii)

115. Which of the following does **not** include in integrated pest management programme?
(1) Use of pest resistant or tolerant varieties
(2) Controlled use of pesticides
(3) Complete destruction of those plants species which are food for pests
(4) Use of repellents

116. How many of the following are biofertilisers that can fix both carbon dioxide and atmospheric nitrogen?

Rhizobium, Azotobacter, Nostoc, Oscillatoria, Glomus, Azospirillum, Anabaena

(1) Three (2) Four
(3) Two
(4) Five

117. A group of individuals of the same species inhabiting a given area is called
(1) Ecosystem
(2) Biome
(3) Population
(4) Habitat

118. The subject which studies the interactions among organisms and between the organisms and its physical environment is called
(1) Ecology
(2) Ethology
(3) Aetiology
(4) Anemology

119. The biome that can receive mean annual rainfall upto 400 cm is
(1) Grassland
(2) Temperate forest
(3) Coniferous forest
(4) Tropical forest

120. Topographic factor of environment includes
(1) Temperature, water and wind
(2) Composition of soil
(3) Slope, altitude and surface behaviour of earth
(4) Types of vegetation and diversity of animals

121. Salanity of inland water in parts per thousand is
(1) Less than 5
(2) 25-40
(3) 50-80
(4) More than 100

122. Read the following statements and choose the option which is true for them.

Statement-1 : Each individuals of a species in a population has a distinct niche.

Statement-2 : Two individual of different species do not occupy exactly the same niche.

(1) Both of the statements are correct
(2) Both of the statements are incorrect
(3) Only statement-1 is correct
(4) Only statement-2 is correct

123. Choose the option which is **incorrect** regarding stenothermal organisms.
(1) These organisms are restricted to narrow range of temperatures
(2) Vast majority of organisms belong to this category
(3) No mammals are stenothermal
(4) Some of these organisms are found in very cold areas of earth

124. The main reason of shifting of climate zones on earth is
(1) Rotation of earth on its axis
(2) Change in the types of vegetation in different areas of earth
(3) Gradual increase in average global temperature
(4) Revolution of earth around the sun

125. Many animals use photoperiod and change in light intensity as cues for
(a) Reproductive activity
(b) Foraging activity
(c) Migratory active
The **correct** one(s) is/are
(1) Only (a) (2) Only (b) and (c)
(3) All (a), (b) and (c) (4) Only (c)

126. Regulators are those organisms which
(1) Can change their body temperature with the ambient temperature
(2) Are able to maintain homeostasis by physiological as well as behavioural means
(3) Cannot maintain a constant internal environment
(4) Live only in a narrow geographical regions

127. Very small animals are rarely found in polar regions because
(1) They cannot maintain the body temperature due to large surface area relative to their volume
(2) They are unable to generate heat as small animals are cold blooded
(3) They cannot migrate from one place to other due to small body size
(4) They are unable to store large amount of food in their body due to complex body organisation

128. In organisms, all of the following can be the ways to avoid environmental stress by escaping in time, **except**
(1) Hibernation
(2) Migration
(3) Aestivation
(4) Diapause

129. One of the ways to compensate low oxygen availability at higher altitude by our body is to decrease the
(1) Production of red blood cells
(2) Binding affinity of hemoglobin
(3) Breathing rate
(4) Heart beat rate

130. One year ago there were some fishes in a pond. During the current year, 300 fishes were born and 120 were died. Now, there are 780 fishes in this pond. What would be the birth rate (individuals per fish per year) in this fish population?
(1) 0.23
(2) 0.39
(3) 0.30
(4) 0.50

131. Population age pyramid will be bell-shaped when
(1) The growth rate of population is almost zero
(2) Smaller number of pre-reproductive individuals is followed by larger number of reproductive individuals
(3) There is a very high proportion of pre-reproductive individuals
(4) The number of post-reproductive individuals are more than that of pre-reproductive individuals

132. Which of the following **cannot** said to be a morphological adaptation characteristic of a desert plant?
(1) Cuticle on their leaf surface
(2) Sunken stomata
(3) Occurance of CAM pathway
(4) Leaves reduced to spines

133. Which reward is provided by the plants for their seed dispersers?
(1) Pollen
(2) Nectar
(3) Green leaves
(4) Nutritious fruits

134. The interaction of two species in which one is benefitted and other is neither harmed nor benefitted is called
(1) Protocooperation
(2) Predation
(3) Commensalism
(4) Competition

135. One of the examples of amensalism is the interaction between
(1) Crocodile and plover bird
(2) *Penicillium* and bacteria
(3) Clown fish and sea anemone
(4) *Cuscuta* and hedge plant

ZOOLOGY

136. Which of the following is considered as primary lymphoid organ?
(1) Bone marrow
(2) Spleen
(3) Lymph node
(4) Tonsil

137. Select **mismatch** w.r.t. diseases and their mode of transmission.

(1)	Amoebiasis	–	Faeco-oral route
(2)	Malaria	–	Biting of female *Anopheles* mosquito
(3)	Ascariasis	–	Biting of female *Culex* mosquito
(4)	Pneumonia	–	Inhaling droplets/Aerosols

138. *Plasmodium falciparum* is a causative agent of a vector-born disease called
(1) Elephantiasis
(2) Malignant malaria
(3) Benign malaria
(4) Yellow fever

139. HIV is classified in category of
(1) Flavi-ribo viruses
(2) Retroviruses
(3) Paramyxoviruses
(4) Influenza viruses

140. The following are some well known abbreviations, which are used for human diseases / organisation / immunity. Choose the option having **incorrect** abbreviation from them.
(1) MALT – Mucosa - associated lymphoid tissue
(2) CMI – Cell-mediated immunity
(3) AIDS – Acquired immuno deficiency syndrome
(4) NACO – Narcotics analysis control organisation

141. All of the following are features of innate immunity **except**
(1) It is present since birth in organisms of a particular species
(2) It prevents invasion of micro-organisms into our body
(3) Physical and physiological barriers are two different types of innate immunity
(4) It keeps the memory of the first encounter of any pathogen

142. IgG being a smallest antibody can cross placenta to provide which type of immunity to foetus?
(1) Natural active immunity
(2) Natural passive immunity
(3) Artificial active immunity
(4) Artificial passive immunity

143. In an antibody, antigen binding site is present on
(1) Only N-terminal of heavy chain
(2) Only C-terminal of light chain
(3) N-terminals of both heavy and light chains
(4) C-terminals of both heavy and light chains

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144. Match the following column-A with column-B and choose the option having only **correct** match.

	Column-A		**Column-B**
A	Physical barriers	p	Acid in the stomach
B	Physiological barriers	q	Skin on our body
C	Cellular barriers	r	Interferons
D	Cytokine barriers	s	Natural killer cells

(1) A(p), B(q), C(r), D(s)
(2) A(s), B(r), C(q), D(p)
(3) A(q), B(p), C(s), D(r)
(4) A(r), B(s), C(p), D(q)

145. All of the following symptoms are associated with amoebic dysentery **except**
(1) Constipation
(2) Abdominal pain and cramps
(3) Stools with excess mucus and blood clots
(4) Frequent vomiting which leads to acidosis

146. Which of the following is **not** included in animal husbandry?
(1) Raising livestock
(2) Caring of livestock
(3) Breeding of livestock
(4) Slaughtering of livestock

147. Which of the following option is **not** the aim of animal breeding?
(1) Increasing the yield of animals
(2) Improving the desirable qualities of the product
(3) Improving resistance of various diseases
(4) Decreasing growth rate of livestock

148. In dairy farms, milk yield is primarily dependent on the quality of
(1) Breeds in the dairy farm
(2) Feeding of cattles
(3) Shelter of cattles
(4) Hygiene of handlers

149. Continued inbreeding especially close inbreeding, usually does **not** increase
(1) Infertility
(2) Inbreeding depression
(3) Homozygosity
(4) Productivity

150. Mule is an example of
(1) Inbreeding
(2) Out-crossing
(3) Cross-breeding
(4) Interspecific hybridisation

151. Houseflies act as mechanical carrier for disease
(1) Malaria (2) Filariasis
(3) Amoebiasis (4) Hepatitis-B

152. Select the **mismatch** w.r.t. *Plasmodium*.

(1)	Sporozoite	–	Infective stage for humans
(2)	Haemozoin	–	Toxic substance
(3)	Gametocytes	–	Sexual stages develop in RBCs
(4)	Gametogenesis	–	Starts within salivary glands of female *Anopheles* mosquito

153. Complete the analogy.
Ringworms : Microsporum : : Pneumonia : ____
(1) *Haemophilus influenzae*
(2) *Staphylococcus*
(3) *Salmonella typhi*
(4) *Trichophyton*

154. Read the following statements A and B and choose the **correct** answer.

Statement-A : Only B-lymphocytes are responsible for both primary and secondary immune responses in our body.

Statement-B : After first primary response **s**ubsequent encounter with the same pathogen elicits a highly intensified amnestic response.

(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is incorrect
(4) Only statement B is incorrect

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155. Probability of Graft-rejection is maximum in grafting between organisms of two different species.

Which type of immune response is mainly responsible for such graft rejection?

(1) Autoimmune response

(2) Cellular barrier immune response

(3) Cell-mediated immune response

(4) Non-specific innate immune response

156. Read the following statements w.r.t. AIDS and choose the option with only **correct** statements.

(a) HIV is an enveloped diploid virus containing one molecule of single stranded RNA and causes AIDS.

(b) HIV infected mother can't transfer AIDS to their foetus

(c) AIDS is a curable disease when diagnosed earlier and treated property

(d) NACO educate peoples about AIDS.

(1) (a), (b), (c) and (d)

(2) (b) and (d)

(3) (d) only

(4) (a), (b) and (c)

157. All of the following are general causes of transmission of HIV infections **except**

(1) Sexual contact with infected person

(2) By transfusion of contaminated blood

(3) From infected mother to her child through placenta

(4) By shaking hands with HIV infected person

158. Both B and T lymphocytes of mammalian immune system are mainly produced from stem cells which are present in

(1) Spleen (2) Thymus

(3) Bone marrow (4) Lymph node

159. Which of the following antibodies are mainly involved in primary and secondary immune response respectively?

(1) IgM and IgG

(2) IgG and IgM

(3) IgA and IgD

(4) IgA and IgE

160. It is estimated that A and B cover more than 70% of the world's livestock population but contribute only 25% of the world farms production. Choose the option which **correctly** fills the blanks A and B.

	A	B
(1)	USA	China
(2)	India	Brazil
(3)	India	China
(4)	China	Japan

161. Rearing of honeybees for obtaining honey and bee wax is called

(1) Pisciculture (2) Sericulture

(3) Apiculture (4) Aquaculture

162. Bird flu is caused by

(1) Fungus

(2) Bacteria

(3) Protozoan

(4) Virus

163. Keeping beehives in crop field during flowering period increases all of the following **except**

(1) Honey yield

(2) Wax yield

(3) Crop yield

(4) Pollination in wheat

164. Read the following statements w.r.t. poultry farm management.

A. Selection of disease free and suitable breeds.

B. Proper and safe condition of the farm

C. Proper hygiene and health care with proper food without water.

D. Temperature of poultry farm should be higher than optimum for more egg laying.

Choose the option which represents all true (T) and false (F) statements **correctly.**

	A	B	C	D
(1)	T	F	T	F
(2)	T	T	F	F
(3)	T	F	T	T
(4)	F	F	T	T

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165. Cancer cells differ from normal cells because they are

(1) Dividing cells with low telomerase activity

(2) Undergoing rapid cell division and exhibit metastasis

(3) Well differentiated cells than normal cell

(4) Non-dividing cells of the body

166. Read the following statements A and B carefully and choose the **correct** answer

A. Morphine and marijuana are used clinically as analgesics.

B. Both morphine and Marijuana have their receptors in brain, for their interaction to relieve pain.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

167. Match the following column I and column II.

Choose the option with only **correct** match.

	Column-I		**Column-II**
a.	Codeine	(i)	*Claviceps purpurea*
b.	Charas	(ii)	*Erythroxylum coca*
c.	Cocaine	(iii)	*Cannabis sativa*
d.	LSD	(iv)	*Papaver somniferum*

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(iii), b(iv), c(i), d(ii)

168. Drugs obtained from plants like *Atropa belladona* and *Datura* belong to which of the following category?

(1) Hallucinogen (2) Stimulants

(3) Sedatives (4) Pain-killers

169. Chronic cigarette smoking causes damage to alveolar walls to form large air sacs which decreases total surface area for gaseous exchange. This disease is called

(1) Asthma (2) Silicosis

(3) Bronchitis (4) Emphysema

170. Which of the following alkaloid interferes with the transport of the inhibitory neuro-transmitter dopamine?

(1) Morphine (2) Diacetylmorphine

(3) Cocaine (4) Atropine

171. If regular dose of drugs/alcohol abuse is abruptly discontinued, it leads to withdrawal syndrome characterised by all of the following symptoms, **except**

(1) Anxiety (2) Respiratory failure

(3) Nausea (4) Sweating

172. The most feared property of malignant tumors is

(1) Starved to mutations

(2) Uncontrolled multiplication

(3) Competing for vital nutrients

(4) Metastasis

173. Which of the following drug is **not** used for treatment of allergy?

(1) Anti-histamine (2) Adrenaline

(3) Cortisol (4) Serotonin

174. Side effects of anabolic steroids are listed below. Choose the option which represents side effects of these drugs only in females.

I. Masculinisation II. Agressiveness

III. Mood swings IV. Breast enlargement

V. Deepening of voice

(1) I and V (2) II, III and IV

(3) I, II and V (4) I, II, III, IV and V

175. Select the **incorrect** statement.

(1) Nicotine stimulates adrenal medulla to release adrenaline and non-adrenaline into blood, which increase heart rate and blood pressure

(2) Smoking increases carbon monoxide content in blood and reduces the concentration of haembound oxygen

(3) Smoking is associated with increased incidence of cancer of liver

(4) Any addict requires counselling and medical help to get rid of the habit

176. When breeding is between two different breeds, it is called

(1) Inbreeding

(2) Out-crossing

(3) Cross-breeding

(4) Interspecific hybridization

177. Which of the following fishes are marine edible fishes?

(1) *Catla*, Rohu (2) Sardines, pomfrets

(3) *Labeo*, pomfrets (4) Rohu, mackerel

178. MOET is a programme for herd improvement which takes place in following 6 steps.

A. A cow is administered with FSH – like hormone

B. Animal is artificially inseminated

C. Follicle maturation and superovulation

D. Fertilized eggs are transferred to surrogate mother

E. Eggs are recovered non-surgically

F. Genetic mother is free for another round of superovulation

Choose the option which arrange these steps in **correct** order starting from beginning.

(1) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

(2) $A \rightarrow C \rightarrow B \rightarrow E \rightarrow D \rightarrow F$

(3) $A \rightarrow B \rightarrow D \rightarrow C \rightarrow F \rightarrow E$

(4) $A \rightarrow C \rightarrow D \rightarrow F \rightarrow E \rightarrow B$

179. 'Blue revolution' is related with

(1) Fishery industry

(2) Dairy farms

(3) Poultry farms

(4) Sericulture

180. When macrophages is infected by HIV, viral A is introduced into cell and viral B incorporates into host genome. Here A and B represents

(1) A-RNA, B-DNA

(2) A-DNA, B-RNA

(3) A-Capsid, B-Enzyme

(4) A-Virion, B-Genome

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